Collective Seed Storage: Share a Tiny Space of Your Freezer to Increase Resilience of Ex-situ Seed Conservation

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***Abstract*—More than 95% of the crop genetic erosion articles analyzed in [1] reported changes in diversity, with nearly 80% providing evidence of loss. The lack of diversity presents a severe risk to the security of global food systems. Without seed diversity, it is difficult for plants to adapt to pests, diseases, and changing climate conditions. Genebanks, such as the Svalbard Global Seed Vault, are valuable initiatives to preserve seed diversity in a single secure and safe place. However, according to our analysis of the data available in the Seed Portal, the redundancy for some species might be limited, posing a potential threat to their future availability. Interestingly, the conditions to properly store seeds in genebanks, are the ones available in the freezers of our homes. This paper lays out a vision for Collective Seed Storage relying on a peer-to-peer infrastructure of domestic freezers to increase the overall availability of seeds. We present a Proof-of-Concept focused on monitoring the proper seed storing conditions and incentive user participation through a Blockchain lottery. The PoC proves the feasibility of the proposed approach and outlines the main technical issues that still need to be efficiently solved to implement the proposed vision.**

I. Introduction

There are over 50,000 edible plants worldwide, but just 15 provide 90% of the world's food energy intake and among these, rice, corn (maize), and wheat contribute up to two-thirds [1]. We have to put all our efforts to preserve those fundamental energy assets.

The need for seed diversity A key objective in preserving these plants is to protect seed diversity, that is, the wide range of plant species, cultivars and genetic materials found within a given crop. This genetic diversity is a vital resource on which the future of our food systems depends.

As climate change intensifies, bringing extreme weather events along with emerging pests and diseases, the presence of specific genes becomes critical. These genes enable certain crop varieties to withstand heat, drought, pests, and pathogens, forming the foundation of agricultural resilience and adaptation.

A limited seed pool not only threatens our ability to cope with environmental stresses but also risks the loss of flavor, nutritional value, and the cultural heritage embedded in traditional cuisines. Genetic diversity improves agriculture's ability to adapt to changing climates, regional growing conditions, and evolving consumer preferences.

[1]https://education.nationalgeographic.org/resource/food-staple/

Without broad access to diverse crop varieties, future generations will face serious constraints in shaping agriculture to meet their needs and respond to environmental challenges.

Our preservation efforts should also extend to Crop Wild Relatives (CWRs), as they play an important role. They are wild plant species that are closely related to cultivated crops and can provide genetic diversity that may not be available in current cultivated varieties. This diversity may be used to improve the productivity, resilience, and quality of agricultural products. The Earth is losing plant genetic diversity at an unprecedented rate. More than 95% of the crop genetic erosion articles analyzed in [1] reported changes in diversity, with nearly 80% providing evidence of loss. The extent of this loss varied depending on species, taxonomic and geographic scale, region, and analytical approach. In many countries, seed diversity has been irreversibly diminished, limiting farmers' crop options.

Efforts to preserve seed diversity. Preservation of seed diversity is considered a common fundamental goal among the countries organized in the Multilateral System of Access and Benefit Sharing (MLS [2]), part of the International Treaty on Plant Genetic Resources for Food and Agriculture (PGRFA). MLS allows member states to share and exchange plant genetic resources for research, breeding, conservation, and training. Recipients sign a Standard Material Transfer Agreement (SMTA) [3] that specifies conditions for use and benefit sharing. MLS includes 64 of the world's most important crops that together represent 80% of all human consumption derived from plants [4].

The main strategies to preserve seed diversity are conservation *in situ* and *ex situ* .

In situ conservation relies on the conservation of seeds and plants in their natural or traditional environment, such as farms where traditional varieties (landraces) are maintained or in the wild where CWR grow. Participatory seed breeding and community seed systems engage local farmers in storing, conserving, sharing, and improving diverse crop varieties adapted to local conditions. Seed guardians (or seed custodians), who may be smallholder farmers, indigenous communities, or members of community seed banks, such as [5]–[8], actively preserve, cultivate, and exchange traditional or locally adapted varieties, often operating outside formal industrial seed systems.

Criteria	In Situ Conservation	Ex Situ Conservation
Location	Natural environment / farms	Seed banks, seed vaults, labs
Evolution	Plants continue to adapt naturally.	Static; no natural evolution
Risk of Loss	Higher (e.g., land-use change, climate)	Lower (secure and controlled conditions)
Cultural Link	Strong; maintains traditional knowledge	Weak or absent
Access & Sharing	Local and informal	Global access via formal mechanisms (e.g., SMTA)
Documentation	Often limited or informal	Detailed and standardized
Cost	Lower short-term cost	Higher cost (infrastructure and maintenance)
Best Use	Wild relatives, landraces	Major crops, backup, and global breeding

TABLE I
COMPARISON BETWEEN IN SITU AND EX SITU SEED CONSERVATION METHODS.

Although in situ conservation has the merit of actively participating local communities and farmers, thus contributing to the conservation of not only the seeds, but also the traditions and skills to take care of them, the access to the material and the documentation is fragmented and difficult to handle. These problems have led to the development of several more structured ex situ conservation initiatives, where seeds are preserved outside their natural habitat in the so-called *seed banks*.

Seed banks. A seed bank is a gene bank, that stores seeds to preserve genetic diversity, namely the genes that make each plant variety unique. It ensures that this genetic heritage is safely conserved and available for people to use.

The Millennium Seed Bank Partnership (MSB) [9] is the largest ex situ plant conservation program in the world, mainly focused on wild species. MSB focuses on the conservation of as many species as possible (interspecific diversity) prioritizing those that are endangered, endemic, or economically important. It provides seeds to registered organizations for research, restoration, and reintroduction purposes.

Crop Trust [10] is an international organization focused on building and supporting a global system of seed banks to preserve the genetic diversity of food crops, especially cultivated varieties and landraces. It provides long-term grants and partnership agreements to several international seed banks, particularly those within the Consultative Group on International Agricultural Research (CGIAR) system. CGIAR genebanks [11] manage collections of more than 20 staple crops in 10 locations across five continents to distribute seeds for research, training, and breeding under the MLS framework.

Genesys is an online platform[2] managed by Crop Trust that provides access to information about plant genetic resources conserved in genebanks around the world. It serves as a global catalogue, aggregating data on millions of accessions from over 450 genebanks, included CGIAR ones.

The Crop Trust also co-manages the Svalbard Global Seed Vault in Norway (SGSV)[3], which serves as a global backup facility for seed banks worldwide [12]. Its purpose is to provide long-term backing for genebank collections to secure the foundation of our future food supply against threats such as mismanagement, accidents, equipment failures, funding cuts, war, sabotage, disease, and natural disasters. In 2019, seeds were returned from Svalbard to restock an international genebank destroyed by the war in Syria.

Contrary to MSB it is not open to researchers and the depositors are the only ones who can withdraw their own seeds. The focus of SGSV is to preserve as much diversity as possible within crop species and their wild relatives (intraspecific diversity).

Seed banks must ensure that their holdings remain alive. They periodically check the viability of the samples, namely their ability to grow into productive plants and produce fresh material when needed. However, the fundamental strategy to preserve such diversity is the availability of good backup systems, namely other seed banks where duplicate samples are delivered for safe keeping.

Seed preservation pipeline The standards for seed preservation [13]–[15] recommend implementing a comprehensive preservation pipeline, summarized in Table II. The underlying principle is that safe storage alone is insufficient; it is equally essential to establish the conditions and maintain the knowledge necessary to reliably propagate seeds over time. In this paper, however, we focus exclusively on the second step of the pipeline, motivated by the open problem discussed in the next section. In Section V, we explain why, despite this limitation, we consider this step a fundamental contribution toward the eventual implementation of the full preservation pipeline in a fully distributed setting, encouraging citizen participation.

Open problem The SGSV is unanimously considered the backup seed facility that provides the highest standards of safety and security. The location where SGSV is built lacks tectonic activity and has permafrost, which aids in preservation; furthermore, it is 130 m above sea level, keeping the site dry even if the ice caps melt. The refrigeration units also cool the seeds to -18,-20 °C, the internationally recommended standard temperature [13]. If the equipment fails, it is estimated that it will take two centuries for it to warm to 0 °C.

The 2023 SGSV Annual Progress Report [16] reveals that, 1267127 samples deposited by more than 100 genebanks/institutes and representing more than 6,000 plant species are stored in the facility. The SGSV now safeguards seeds of more than 250,000 types of wheat, 160,000 types of rice and 46,000 types of maize, the fundamental sources of food energy intake.

Figure 1 shows the distribution of the depositors (i.e.

[2]https://www.genesys-pgr.org/
[3]https://www.seedvault.no/

Step	Description
Acquisition of germplasm	Collecting seeds or other reproductive material from diverse sources, including wild relatives, local varieties, and breeding programs, to ensure genetic diversity.
Drying and storage	Reducing seed moisture content and storing under controlled conditions (temperature, humidity) to maximize longevity and maintain viability.
Seed viability monitoring	Periodically testing seeds to ensure they remain capable of germination and identifying when regeneration is needed.
Regeneration	Growing out seeds to produce fresh seeds, maintaining genetic integrity while replacing aging or depleted stock.
Characterization	Documenting observable traits (morphology, growth habits, resistance traits) to describe the germplasm and facilitate its use.
Evaluation	Assessing performance under different conditions, including agronomic, nutritional, or stress-tolerance traits, to determine the value of the germplasm.
Documentation	Recording all relevant data on origin, traits, storage conditions, and usage to ensure traceability and informed utilization.
Distribution and exchange	Providing seeds to breeders, researchers, farmers, or other genebanks to promote use, collaboration, and conservation.
Safety duplication	Creating backup copies of germplasm, often in separate locations or facilities, to protect against loss from disasters or failures.
Security and personnel	Security and personnel – Ensuring the physical, legal, and operational security of the seed collection, and maintaining trained staff to manage the conservation process.

TABLE II
STEPS IN THE SEED CONSERVATION PIPELINE

genebanks and institutes) for the species obtained by analyzing the data from the SGSV Seed Portal [4].

The vast majority of species have a single depositor. In other words, the data show that the samples are available at SGCV and at the depositor only. As an example, Oryza sativa (i.e. the Asian cultivated rice) is the most common rice cultivated as a cereal. The result of the query to the Seed Portal with the keyword *Species: Oryza sativa* is 29 depositors, 171193 accessions and 136 countries of collection. The same query with the keyword *Species: Hygroryza aristata*, gives 1 depositor (i.e. the Rice Research Institute), 4 accessions and 1 country of collection as also confirmed by the Genesys online platform. Rice is currently by far more "important" than Hygroryza for food supply, however, the Hygroryza is a CWR

[4]https://seedvault.nordgen.org/Search

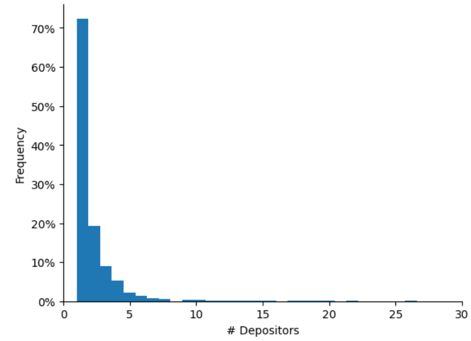

Fig. 1. The distribution of the depositors for the species in the SGSV. The majority have a single depositor posing a potential threat to seed availability.

of rice [17] and, as already observed, it might contribute to the development of more resilient new rice varieties.

The low level of redundancy in the availability of seeds samples backups for the majority of the species, might have serious consequences; even in the best genebanks, things can occasionally go disastrously wrong. The question we aim to investigate in this paper is whether it is possible to further increase the overall availability of seeds by a robust and resilient backup infrastructure capable of preserving seed diversity with limited investments.

Contributions of the paper. We recognize that a comprehensive strategy for seed preservation must address all components of the conservation pipeline skteched in table II and should actively encourage citizen participation to raise awareness and engagement. Nonetheless, having a robust and resilient backup remains a fundamental element of such a strategy. In this paper, we focus specifically on this essential, though narrowly defined, objective.

Interestingly, the temperature conditions in domestic freezers can meet the international standard for long-term seed conservation [13], [14]. The main idea of this paper is to develop a system inspired by the principles of the shared economy, capable of safely storing the seeds in our home freezers, thus contributing to the development of a Collective Seed Storage (CSS) that can increase the availability of seeds and the overall resilience of the world's crop diversity. To our knowledge, this paper is the first attempt to build a peer-to-peer infrastructure to preserve seed diversity. We first identified a set of requirements, and then we built a proof-of-concept of the CSS that proves the technical feasibility of two fundamental tasks, namely the IoT monitoring activity of seed conservation conditions in domestic freezers and the interaction of the IoT node with a Blockchain smart contract to autonomously manage a lottery to incentivize user participation. The development of this PoC, allows us to identify the main issues that need to be investigated in future work.

II. Background and State of the Art

Moisture, temperature, and the proportion of oxygen are key environmental factors that affect seed deterioration and loss of viability [18]. Reducing seed moisture content (MC), namely

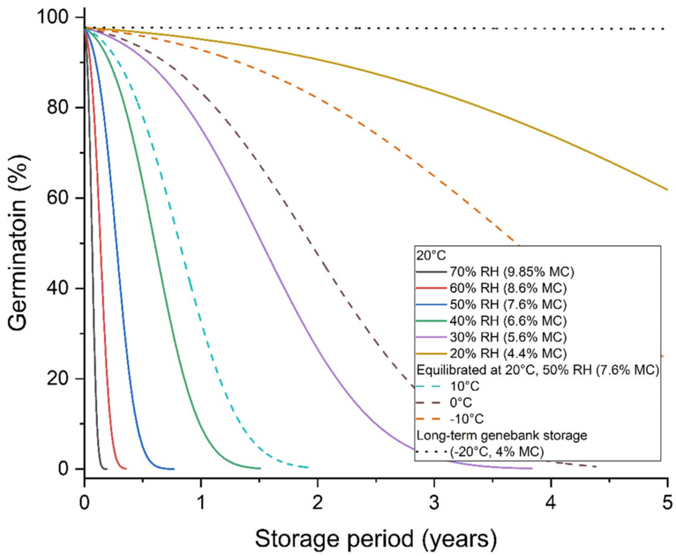

Fig. 2. Figure 2 from [18] Creative Commons CC BY license. Predicted survival curves for seeds of foxglove at different conditions

the amount of water in the seed, to certain thresholds increases longevity predictably for approximately 90% of species. These species are classified as being *orthodox* in their seed storage requirements, and generally retain viability and germinability even after storage for long periods under suitably dry, cool conditions. In general, most of our food energy intake from plants is orthodox. Examples of crop plants that bear orthodox seeds are wheat (*Triticum aestivum L.*), cereal rye (*Secale cereale L.*), and legumes such as the common bean (*Phaseolus vulgaris L.*), rice (*Oryza sativa*) and in general most grains and legume types, namely most of our food energy intake from plants.

The Seed Information Database[5] provides a tool to Predict Storage Time for orthodox seeds according to the equations of Ellis and Roberts [19]. Figure 2 is taken from [18] and shows the predicted survival curves for seeds of foxglove (Digitalis purpurea L.) at different conditions of temperature, relative humidity (RH) and moisture content (MC). The figure clearly shows that the long-term genebank storage conditions, namely -20°C and 4% seed moisture content, allow us to preserve germination for years. The Food and Agriculture Organization (FAO) standardized the process for long-term storing of orthodox seeds in [13], [14]. We require our system to comply with these standards, as specified in **R1**, section III.

In this paper, we share the definition of shared economy by Frenken et al. *"consumers granting each other temporary access to under-utilized physical assets (*idle capacity*), possibly for money."* [20].

Example of shared economy ranges from carpooling (e.g. BlaBlaCar), to share rooms in an apartment (e.g. Airbnb) or to utilize transport capacity that may be wasted in inefficient transport operations (e.g. Shiply), just to mention few examples. In our case, we share the currently idle capacity of a tiny portion of the freezer, sufficient to store the seeds.

The sharing economy can benefit from integration with the Internet of Things (IoT) [21]. IoT technologies can be used to monitor the availability of idle capacity and to verify that specific requirements are met (see, for example, requirement **R1** in Section III). In the scenario we envision, however, IoT is not merely a means of monitoring proper conservation conditions; it also serves as a tool for sharing objective, verifiable knowledge among participants. In this way, IoT contributes to the collective knowledge base required to fully implement the seed preservation pipeline introduced in Section I.

A vast body of literature explores the data economy and shared-resource paradigms. For instance, in [22], the authors present the design, implementation, and evaluation of an ambient environment-aware system that supports fine grained urban air quality monitoring through mobile crowdsensing integrated into a bike sharing infrastructure.

SETI@home[6] and Folding@home[7] are two prominent examples of the sharing economy in which the idle capacity, namely, the computational power of users' personal computers, is used to support large scale distributed computations, such as radio signal analysis and protein dynamics simulation.

Unlike the aforementioned examples, although the freezer space required is limited, the seeds are expected to be stored for extended periods, potentially spanning several years. While participation in a global effort for seed preservation should not necessarily require compensation, the long-term nature of seed storage may nonetheless limit user participation. This, in turn, calls for the definition of appropriate incentive mechanisms to encourage the involvement of a sufficiently large number of participants.

As discussed in [23], enjoyment, economic incentives, reputation, and self-fulfillment are among the primary motivations for participation in sharing-economy initiatives. In this context, we are particularly interested in investigating how blockchain technology can be leveraged to distribute and manage such incentives.

In [24], the authors explore blockchain-based incentive mechanisms in which participation in a lottery serves as the reward: only peers actively contributing to the CSS are eligible to purchase tickets and potentially win. This approach illustrates how decentralized technologies can be used to align individual incentives with collective goals.

III. Requirements

In this section we describe the main requirements that drive the design of the CSS proof-of-concept.

R1 Storing conditions of seeds in domestic freezers must comply with [13], [14]. In this paper, we focus on long-term storage of orthodox seeds. Specifically we assume that after drying, samples meant for long-term storage are packaged under controlled conditions, in clearly

[5]https://ser-sid.org/viability/storage-time

[6]https://setiathome.berkeley.edu/

[7]https://foldingathome.org/

labelled airtight containers (Standard 4.2.2 [13]) and then samples are ideally stored at -18°C ± 3 percent and relative humidity of 15 ± 3 percent (Standard 4.2.3 [13]). Figure 4 of [14] is a summary diagram of the workflow and activities for drying and storage and clarifies that the use of subzero freezers is acceptable if ideal conditions are not available.

The success of this project relies on the active and numerous participation of users. To encourage the participation, the system should:

R2 Simplify the installation without requiring any specific technical skill or infrastructure. The installation of the system should be user friendly and based on off-the-shelves equipment. We only assume users have a WiFi at home.

R3 Not require unnecessary cables which complicates and limit the installation options.

R4 Run for at least 1 year without the need of intervention by the user.

R5 Support the implementation of suitable incentive mechanisms.

IV. Proof-of-concept

The main purpose of this section is to demonstrate the feasibility of the proposed approach by the implementation of a Proof-of-Concept (PoC) of the Collective Seed Storage driven by the above requirements. The PoC will also outline the main technical problems that still need to be efficiently solved to realize a fully fledged solution.

The airtight container storing the seeds is equipped with a humidity and temperature sensor (the DHT22) connected to an IoT sensor node, in our case the Esp32 Devkit V1 (Esp32 in the following), which implements the logic to acquire the sensor readings and to deliver them to the smart contract in charge of dispensing the incentives to the participants.

The Esp32 delivers the data to the smart contract by the on-board WiFi. WiFi access points are nowadays largely available in homes (see **R2**) making this technology appropriate for the project, furthermore, the nature of the phenomenon to be monitored does not require frequent data acquisition; temperature and humidity in the freezer do not suddenly change. For this reason, the node alternates long period of *deep sleep* with a short active period where two main tasks are performed, namely data acquisition and communication. In *deep sleep* mode, only the RTC Timer and RTC Memory are active, reducing the consumption to 10μA. The resulting low duty cycle prolongs the lifetime of the node, and allows us to meet the requirement **R4** as quantified in section IV-C.

To satisfy **R3**, the Esp32 is powered by a 3.6V LiFePo4. This solution does not require any voltage regulator that would introduce losses. In the future, we plan to design a more efficient custom board without all unnecessary components.

The ESP32 operating temperature range[8] is from -40 to 125 °C, so in principle it can also be placed into the freezer together with the DHT22 temperature and humidity sensor, however, the wifi connectivity could be severely hindered. In our experiments, when the ESP32 is placed into the freezer, more then 40% of the packets do not reach the server. Furthermore, it is well known the the efficiency of batteries decreases significantly in cold temperatures: at -20°C to -40°C, a LiFePo4 battery may only achieve about 60% to 40% of its rated capacity[9]. Finally, despite being small, the ESP32 steal some further space in the user's freezer. For these reasons, the DHT22 is placed into the freezer and it is connected to the ESP32 placed outside, by three tiny wires. This solution is not fully consistent with **R3**, but it is acceptable and the arguments discussed above togheter with the simplicity of the installation, makes it compliant with **R2**.

A sketch of the architecture of the Proof-of-Concept is shown in figure 3. The PoC is build with PlatformIO [10] and the code is available on *https://github.com/andreavitaletti/PlatformIO/tree/main/Projects/web3E_SC*.

In the following, we discuss the implementation of the two main tasks performed by the PoC during the active period, namely monitoring the correct storage of seeds and delivering the observed data to the smart contract in charge of incentive users participation.

A. Monitoring the Correct Storage of Seeds

The temperature and humidity conditions at which seeds are stored in the freezer, are monitored by a DHT22 sensor (see **R1**). It features excellent technical specifications also in terms of power consumption. The sensor can be queried by a single wire protocol: when the sensor node sends the start signal, the DHT22 change from stand-by-mode (40 μA) to measuring-mode (1 mA). The DHT22 will get back to stand-by-mode again upon the completion of the data acquisition.

Figure 4 shows the results of a monitoring activity on a domestic freezer (BOSCH KGN39X23) set at -18 °C of temperature. Samples are taken every 5 minutes. After an initial transient period, the observed temperature oscillates significantly. These conditions do not fulfill the Standard 4.2.3 [13] but are consistent with the acceptable subzero conditions indicated in [14] when the ideal ones are not available. The implications of these conditions in seed conservation and the possibility to reach the ideal ones in domestic freezer are object of future studies.

Domestic freezers are occasionally opened to access the frozen products. We performed a simple experiment to evaluate the impact of this activity on the temperature. We left the freezer close for 1 hour and then we opened it for 30 secs., a more than sufficient interval of time to access the frozen products; the impact on the temperature is marginal at the first decimal figure of the temperature.

[8]https://www.espressif.com/sites/default/files/documentation/esp32_datasheet_en.pdf

[9]https://www.evlithium.com/Blog/lifepo4-battery-temperature-range-capacity-voltage.html

[10]https://platformio.org/

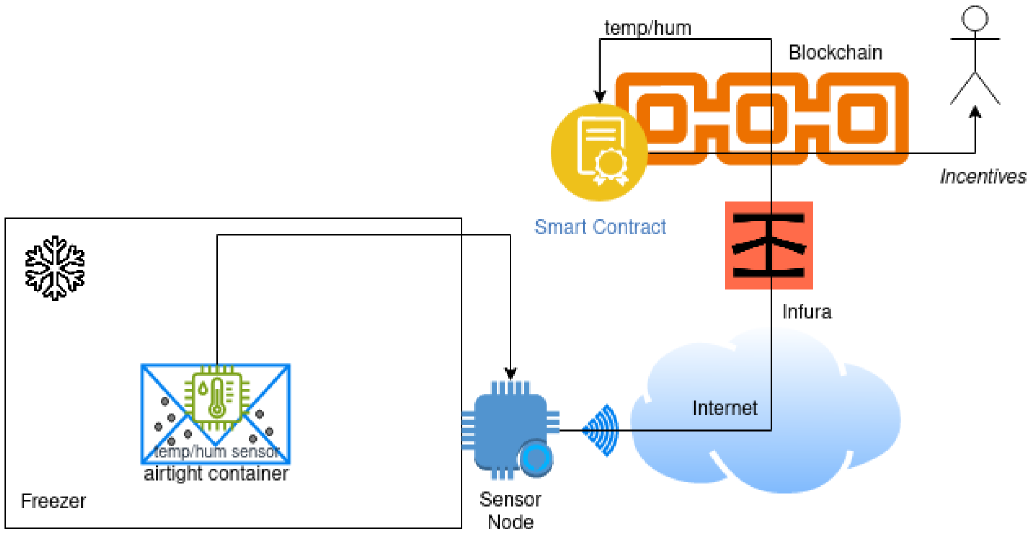

Fig. 3. The architecture of the Proof-of-Concept for the Collective Seed Storage. A temperature and humidity sensor is placed into the freezer and it is connected to an IoT sensor node placed outside. The node communicates the readings on the conservation conditions to a smart contract in charge of dispensing incentives to encourage users' participation.

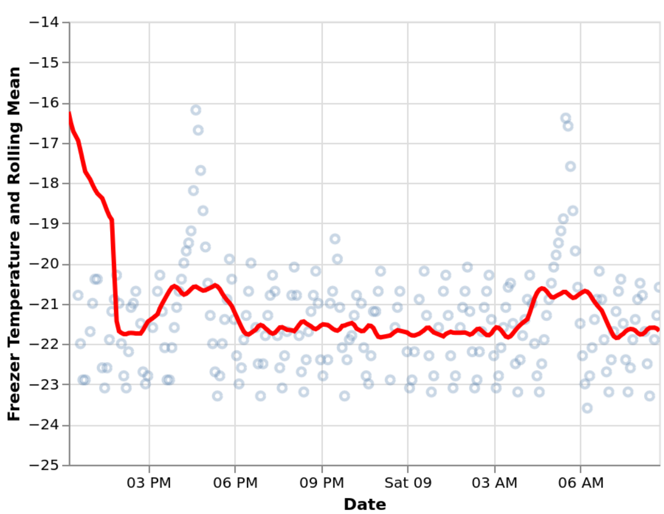

Fig. 4. The observed temperature drops immediately after the placement of the sensor into the freezers, then fluctuates noticeably.

B. Delivering the observed data to the smart contract

Once data have been acquired, they are delivered to the blockchain smart contract in charge of distributing the incentives to encourage users' participation (see **R5**). In our PoC, we envision the development of a blockchain lottery, where users can participate (and possibly have higher chances of winning) if and only if they are active peers of the CSS. The lottery is implemented with a simple smart contract similar to that available at https://github.com/Scofield-Idehen/Lottery_Contract/blob/main/Lot_Contract.sol. The function *enter(uint temp, uint hum)* allows the participation only of active CSS users, namely users that provide data on temperature and humidity in the acceptable range. The function *fund*, allows anyone to fund the lottery. The code of the smart contract, together with some sample transactions, can be accessed at https://sepolia.etherscan.io/address/0x19aeacb63eba19e5e159a5870fa6afce5d3b37ec.

A more complex lottery and a through study on the most suitable incentive mechanisms to be implemented in the smart contract is beyond the scope of this paper and will be investigated in future work.

Most solutions interfacing IoT nodes with smart contracts on the Blockchain, relays on gateways. This would greatly simplify the integration, but we cannot expect that all the houses have the necessary skills and tools to deploy such a relatively complex infrastructure (see **R2**). For this reason, all the code to interact with the smart contract through the Infura API[11] is contained in the firmware of the resource constrained ESP32. This might be a pretty challenging solution due to the limited memory available on the ESP32, indeed our code, based on the WEb3E library [12], occupies 93% of the available flash. This solution foresees only outgoing transmissions (from the home to the Internet), thus increasing the security against possible network threats.

The WEb3E library [13] allows the node to interact with the Solidity smart contract shown in Listing IV-B; it is a simple key-value storage, where the key is the address of the node and value contains the last observed temperature and humidity. The smart contract is deployed on the Sepolia testnet and can be accessed at https://sepolia.etherscan.io/address/0xeda34ef93426dcffde0e9f7c92370d6f8ad9f233 where also the trasactions from the nodes can be inspected.

The implementation required few minor changes to the library, the most important of which is the encoding of the con-

[11]https://www.infura.io/
[12]https://github.com/AlphaWallet/Web3E
[13]https://github.com/AlphaWallet/Web3E

listingA minimalistic Smart Contract to Store/Retrive Sensor Readings In-chain. [linenos,tabsize=2,breaklines,xleftmargin=20pt]soliditysc.sol

tract data instead of relying on the *contract.SetupContractData* function provided by the library; this was necessary to handle some run-time issues with the variadic implementation of the function.

DHT22 readings are given as floating numbers that are not supported by Solidity. To simplify the management of readings by the smart contract also for negative values, we applied the following simple formula $40 + rount(temp)$ with $temp$ ranging from -40°C to 80°C, as defined in the technical specification of the DHT22. Consequently, the value transmitted to the smart contract is a single byte representing a value between 0 (-40°C) and 120 (80°C). The transactions can be accessed at the same URL of the smart contract. Floats can be also encoded in 4 bytes using the IEEE 754 standard, but this makes more complex the logic of the smart contract.

C. Energy Consumption

In this section we report on the experiments to evaluate the energy consumption through an INA219[14]. The Esp32 devkit V1, has an always on red led, with a fix consumption of 10mA. We unsoldered the led and the consumption dropped to less than 4mA, however still pretty far from the nominal $10\mu A$ in deep sleep. In the following, to evaluate the potentials of our work, we assume to achieve the nominal current consumption during the sleeping period.

We sampled several active periods and figure 5 shows the current consumption in one of those periods; similar values have been observed for the others. Active periods are made of two tasks, the *init* task where all the necessary components are initialized, and the *com* where the data are read by the DHT and are transmitted to the smart contract. The *init* task lasts about 2400ms (≈0.0007h) at about 50mA, namely ≈ 0.03mAh, while the *com* task lasts about 3160ms (≈0,0009h) at about about 100mA, namely ≈ 0.09mAh. We assume the duration of the sleep period is X hours at $10\mu A$, namely $0.01X$mAh. The whole consumption for a cycle is $0.03mAh + 0.09mAh + 0.01XmAh$. The ESP32 is powered by a common 3.6v, 1.5Ah LiFePo4 battery. This guarantees a number of cycles $1500mAh/(0.12+0.01X)mAh$. The length of a cycle is $0.0016 + X$ hours. The whole duration in hours will be $1500/(0.12+0.01X)\cdot(0.0016+X)$ and with $X \geq 0.73$ hours, we meet requirement **R4**, namely a duration of 1 year.

Note that the smart contract gets in input an unsigned int made of 32 bytes, so we can pack up to 32 readings (they need 1 byte), into a single call, thus reducing the overall number of required transmissions.

V. Conclusions and Future Work

The pipeline for the conservation of orthodox seeds in seed genebanks [14] involves the steps summarized in table II. Each of these steps is fundamental for a successful preservation effort. In this paper, we concentrate on the storage step. Although it represents only a single element of the complex pipeline, it is nevertheless of fundamental importance. The analysis of the data reveals that even SGSV, the most renowned backup seed facility, has limited redundancy for certain seed samples. This observation forms the basis of our vision for Collective Seed Storage (CSS), a novel shared-economy approach for the long-term storage of seeds in domestic freezers. The CSS concept seeks to support the fundamental goal of preserving seed diversity by leveraging IoT devices to monitor proper storage conditions and blockchain technology to distribute incentives that encourage active user participation.

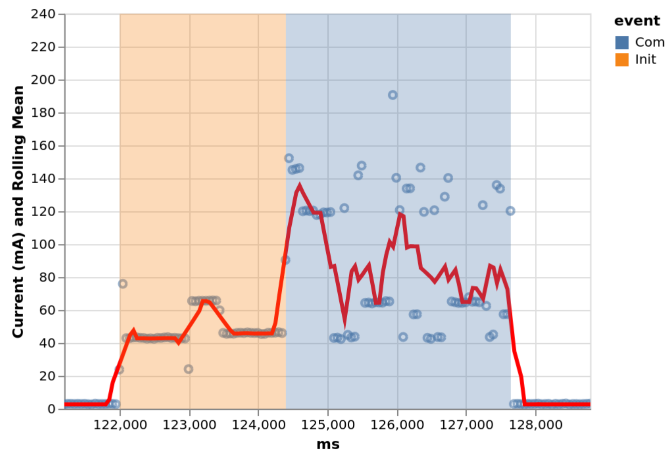

Fig. 5. Sampling current consumption with the INA219

A modern approach to seed preservation should promote more active citizen participation (e.g., Seed Guardians) that extends beyond the mere act of storage. Although the preservation of seeds in domestic freezers is a novel and valuable practice, it may remain, to some extent, a passive activity. However, we argue that the network of IoT devices forming the CSS is not only a powerful tool for monitoring conservation conditions, but also a means to initiate structured knowledge sharing on effective seed preservation. In particular, the availability of quantitative, objective data can support evidence-based practices and serve as a foundation for more active citizen engagement.

Greater involvement, such as participation in seed regeneration and cultivation, would not only directly enhance conservation efforts but also equip participants with practical knowledge of seed growth and management. Although such activities lie beyond the scope of this work, we believe that the sharing-economy principles underlying our CSS concept represent a first step in this direction, laying the groundwork for deeper, hands-on participation in the long-term preservation of seed diversity.

The PoC proves the feasibility of the main technical components of our vision and a future collaboration with fridge producers, might support the development of new products supporting the CSS functionality since the design, also on

[14]https://www.ti.com/lit/ds/symlink/ina219.pdf

industrial refrigerators. The PoC also allowed us to identify the priorities to be investigated in future work as summarized below.

The monitoring activity confirms that acceptable subzero conditions are achievable in domestic freezers; however, to what extent we can better approach the ideal conditions prescribed in Standard 4.2.3 [13] and the implications on seed conservation of the observed temperature fluctuations, need further investigation. While the current board proves the feasibility of the proposed approach, the power consumption can still be optimized and more energy-efficient custom boards, such as the trigboard[15], can significantly improve the performance. The PoC relies on periodic transmission of the observed conditions, the use of an alternative approach based on the communication of abnormal events only, possibly based on Ultra Low Power Coprocessor Programming, should be evaluated.

In this work, user participation is incentivized through a simple lottery. Looking ahead, we aim to explore more sophisticated mechanisms, potentially informed by a game-theoretical framework. Beyond assessing technical feasibility, it is essential to evaluate costs and design strategies that effectively encourage engagement. These strategies could combine advanced tokenomics with the smart monitoring of seeds to deliver value-added services; for instance, alerting users to conditions that may compromise food preservation in their refrigerators. By doing so, the system not only promotes active participation in seed conservation but also provides tangible everyday benefits, bridging the gap between individual action and collective impact.

Facilitating easy and routine access to crop genetic resources is a fundamental goal of the Multilateral System [2]. These resources may include local seed collections preserved in small refrigeration units within research laboratories, national seed banks maintained by government institutions, or comprehensive collections held by international research centers that encompass all known varieties of a given crop. Our goal is to explore how the paper's proposed participative approach can be integrated into the MLS. This includes determining how to regulate the exchanges involving the new category of users sharing their fridges, utilizing the Standard Material Transfer Agreements (SMTAs).

[15]https://trigboard-docs.readthedocs.io/en/latest/

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